

CMR-GAN: Covariance Matrix Reconstruction for Coherent DOA Estimation

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Abstract—Multipath-induced coherent signals cause rank deficiency in covariance matrices, rendering traditional direction-of-arrival (DOA) estimators ineffective. To overcome this limitation, this paper proposes a covariance matrix reconstruction generative adversarial network (CMR-GAN). Distinct from spatial smoothing techniques that sacrifice array aperture, CMR-GAN directly recovers the full-rank covariance matrix without aperture loss. The generator employs a simplified attention-based encoder-decoder architecture to extract robust features, while the discriminator ensures the full rank of the reconstructed matrix for subsequent subspace-based estimation. Simulation results verify that the proposed CMR-GAN significantly outperforms the existing deep learning approach, achieving superior performance and robustness even under challenging conditions.

Index Terms—Direction-of-arrival estimation, generative adversarial network, coherent signal, covariance matrix

I. INTRODUCTION

Direction-of-arrival (DOA) estimation makes a significant contribution in various fields, including radar, detection, navigation, and wireless communications [1], [2]. In ideal scenarios, signals are typically assumed to be uncorrelated, allowing traditional DOA estimation algorithms to achieve satisfactory performance. However, in realistic electromagnetic environments, multipath effects are prevalent, often resulting in coherent signals impinging on the array. This coherence leads to a rank-deficient covariance matrix, bringing about the failure of conventional subspace algorithms such as MUSIC [3] and ESPRIT [4]. Given that accurate coherent DOA estimation significantly advances applications in positioning, radar target detection, and hostile signal identification, research into estimating the DOA for coherent signals is indispensable.

Since signal coherence induces rank deficiency in the covariance matrix, applying appropriate transformations to

restore its full-rank property is a critical precursor to effective DOA estimation. Over the past few decades, spatial smoothing (SS) has been a quintessential approach. By leveraging the cross-covariance information between stationary and moving arrays, [5] introduced a forward/backward convolution kernel algorithm that applies convolutional operations to the signal space matrix, effectively achieving decorrelation and high-resolution estimation. An enhanced spatial smoothing approach was proposed in [6], which flexibly exploits either the covariance matrix of an individual subarray or the cross-covariance matrix derived from distinct subarrays. Another preprocessing method for the covariance matrix is the Toeplitz reconstruction method [7]. This approach designs a specific array configuration to rearrange elements of the covariance matrix into a Toeplitz matrix, thereby achieving full rank. However, the above-mentioned methodologies exhibit certain limitations. Some SS algorithms sacrifice a degree of array aperture, leading to reduced resolution. Conversely, Toeplitz reconstruction tends to result in larger estimation errors. Furthermore, the performance of both methods degrades significantly under circumstances of extreme signal-to-noise ratio (SNR) and limited snapshot numbers.

Over the past decades, the rapid development of deep learning has opened promising avenues for coherent signal DOA estimation [8], [9]. In [10], the authors combined the full-row Toeplitz matrices reconstruction as a preprocessing step with a multi-layer perceptron (MLP). However, this algorithm suffers from large errors at low SNRs. [11] designed a sample matching strategy to estimate coherent DOAs, thereby obviating the necessity for a prior knowledge regarding the source count and bypassing explicit decorrelation preprocessing. In [12], the authors proposed a deep learning method based on the Vision Transformer (ViT) to estimate coherent signals. Although this network captures both detailed and global features, its two-layer ViT encoder design drastically increases computational overhead, making it unsuitable for practical deployment. Au-

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thors in [13] utilized residual network (ResNet) to reconstruct the noise subspace and employed a specific loss function to verify the orthogonality between the noise subspace and the steering vectors. However, this loss function imposes a substantial computational burden during training. The networks designed in [14] and [15] can also handle coherent signals to some extent, but they exhibit suboptimal estimation errors. In summary, existing deep learning methods either perform poorly in harsh environments or require complex network structures and loss functions to enhance training efficacy, indicating that significant defects remain.

To address the aforementioned challenges, this paper proposes the covariance matrix reconstruction generative adversarial network (CMR-GAN) to reconstruct the full-rank property of the covariance matrix. The designed CMR-GAN generator possesses robust feature extraction and decorrelation capabilities, providing a full-rank covariance matrix for the subsequent MUSIC algorithm, while the discriminator is employed to distinguish between generated and real covariance matrices. Furthermore, simulation experiment results attest to the fact that the proposed CMR-GAN obtains high estimated accuracy and robustness in coherent signal DOA estimation.

II. SIGNAL MODEL AND PROBLEM FORMULATION

A. Signal Model

Consider a scenario where K far-field narrowband signals are incident upon a uniform linear array (ULA) comprising M omnidirectional sensors, satisfying $M > K$. The wavelength is denoted by λ , and the spacing between elements d is set to half the wavelength, i.e., $d = \lambda/2$. At time t , the signal vector corresponding to the different incident angles can be defined as $\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_K(t)]^T$. The received signal vector of the array can be represented as

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t) + \mathbf{n}(t) = \sum_{k=1}^K \mathbf{a}(\theta_k) s_k(t) + \mathbf{n}(t), \quad (1)$$

where $\mathbf{n}(t)$ denotes the additive white Gaussian noise vector with zero mean and variance $\sigma^2 \mathbf{I}$, and $\mathbf{A} = [\mathbf{a}(\theta_1), \dots, \mathbf{a}(\theta_K)]$ is the array manifold matrix. The steering vector $\mathbf{a}(\theta_k)$ is expressed as $\mathbf{a}(\theta_k) = [1, e^{-j2\pi d \sin \theta_k / \lambda}, \dots, e^{-j2\pi d (M-1) \sin \theta_k / \lambda}]^T$.

Given that the signals are coherent, we designate $s_1(t)$ as the reference signal. Consequently, the remaining signals can be formulated as scaled versions of the reference signal $s_k(t) = \beta_k s_1(t)$, $k = 1, \dots, K$, where $\beta_k = \eta_k e^{j\phi_k}$ is the complex fading coefficient representing the multipath propagation effect, where η_k stands for the amplitude attenuation factor, and ϕ_k denotes the phase difference.

Based on the aforementioned assumptions, the ideal covariance matrix is given by

$$\mathbf{R} = E[\mathbf{x}(t)\mathbf{x}^H(t)] = \mathbf{A}\mathbf{R}_s\mathbf{A}^H + \sigma^2\mathbf{I}_M, \quad (2)$$

where $(\cdot)^H$ denotes the conjugate transpose operation, $\mathbf{R}_s = E[\mathbf{s}(t)\mathbf{s}^H(t)]$ is the source covariance matrix, and $\mathbf{I}_M$ represents the $M \times M$ identity matrix. In practical scenarios,

however, the ideal covariance matrix is generally inaccessible. Therefore, it is standard practice to estimate it utilizing the sample covariance matrix derived from L snapshots

$$\hat{\mathbf{R}} = \frac{1}{L} \sum_{l=1}^L \mathbf{x}(t_l)\mathbf{x}^H(t_l). \quad (3)$$

B. Problem Formulation

For coherent signals, the covariance matrix $\mathbf{R}_s$ is rank-deficient. This rank deficiency renders subspace-based methods, such as MUSIC and ESPRIT, ineffective for DOA estimation. Consequently, efficient and accurate decorrelation algorithms are required to recover the rank of $\mathbf{R}_s$. In this paper, the decorrelation process is modeled as a neural network.

To alleviate the influence of the noise terms on the rank of the sample covariance matrix and to recover the true rank of the signal subspace, the sample covariance matrix obtained from (3) undergoes sequential noise pre-whitening and normalization. This procedure is expressed as

$$\mathbf{R}_{\text{processed}} = \hat{\mathbf{R}} - \hat{\sigma}^2 \mathbf{I}_M, \quad \bar{\mathbf{R}} = \frac{\mathbf{R}_{\text{processed}}}{\|\mathbf{R}_{\text{processed}}\|_F}, \quad (4)$$

where $\mathbf{R}_{\text{processed}}$ and $\bar{\mathbf{R}}$ represent the matrix after noise prewhitening and the normalized matrix, respectively. $\hat{\sigma}^2$ is the estimated noise power, which is taken as the average of the $M - K$ smallest eigenvalues of $\hat{\mathbf{R}}$, and $\|\cdot\|_F$ stands for the Frobenius norm.

Following this, the processed covariance matrix is decomposed into its real, imaginary, and phase components to accommodate the input requirements of the proposed CMR-GAN. The final input data $\mathbf{X}$ structure can be represented as

$$\mathbf{X} = \text{Concat}(\Re(\bar{\mathbf{R}}), \Im(\bar{\mathbf{R}}), \angle(\bar{\mathbf{R}})) \in \mathbb{R}^{3 \times M \times M}, \quad (5)$$

where $\Re(\cdot)$, $\Im(\cdot)$, $\angle(\cdot)$ denote the operations to extract the real part, imaginary part, and phase angle, respectively. Concatenation performed along the channel axis (dimension 0) is symbolized by $\text{Concat}(\cdot)$.

The problem of recovering the rank of $\mathbf{R}_s$ is addressed by the proposed CMR-GAN, which is similar to the pix2pix generative adversarial network (GAN). The objective of the proposed CMR-GAN is to establish a non-linear mapping to achieve signal decorrelation, defined as

$$\mathbf{R}_{\text{target}} = \mathcal{G}(\mathbf{X}), \quad (6)$$

where $\mathcal{G}(\cdot)$ and $\mathbf{R}_{\text{target}}$ represent the CMR-GAN generator and reconstructed covariance matrix, which is of full rank, respectively.

III. THE PROPOSED CMR-GAN

The architectural topology of the proposed CMR-GAN is illustrated in the following section. The fundamental aim of the proposed framework is to employ a GAN for restoring the rank of the covariance matrix, effectively realizing signal decorrelation. This restoration facilitates the application of subspace algorithms for effectively estimating DOAs. The comprehensive framework of the CMR-GAN, visualized in Fig. 1,

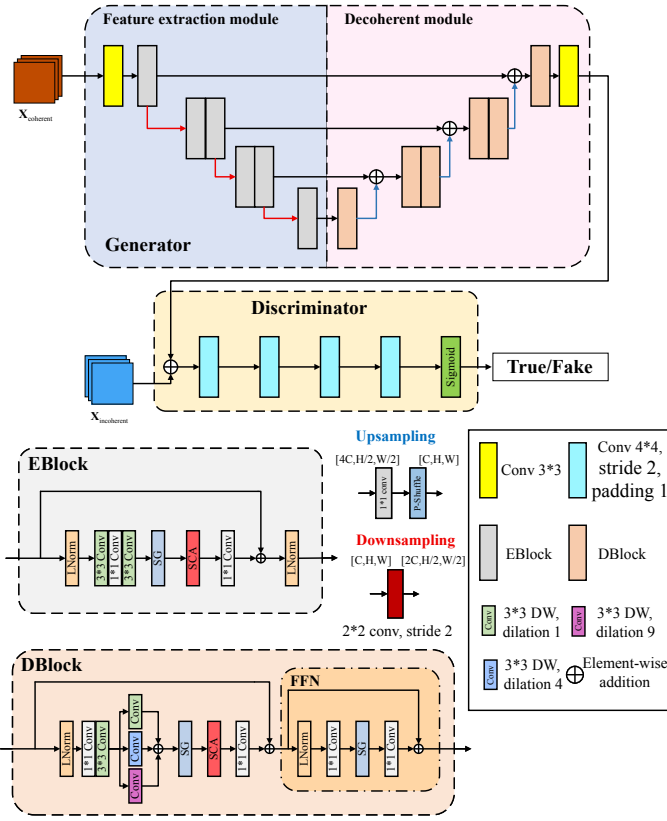


Fig. 1: Proposed CMR-GAN.

integrates a generator with a discriminator. Specifically, the generator is built upon a standard encoder-decoder backbone, which corresponds to the tasks of feature extraction and signal decorrelation, respectively. From a design perspective, both modules maintain a high degree of symmetry. The $\{3, M, M\}$ input data passes through a convolutional layer to increase the channel depth to 32, before proceeding to a series of encoder blocks (EBlock) for feature extraction on the coherent signal covariance matrix. Downsampling is performed between EBlock combinations using 2D convolution operations with a 2×2 kernel and a stride of 2. Furthermore, skip connections are established between each encoder combination and the corresponding decoherent module to prevent the loss of original information.

The structure of a single EBlock is detailed in Fig. 1. In this work, simple gates (SG) and simplified channel attention (SCA) are employed to replace activation functions, thereby mitigating neuron loss. Additionally, layer normalization and residual connections are incorporated to ensure network convergence and prevent the gradient vanishing problem. Ultimately, the feature extraction module provides robust high-level feature representations for the decoherent module.

The decoherent module consists of decoder blocks (DBlock), which share a symmetric structure with the feature extraction module, as shown in Fig. 1. In this module, atrous convolutions with dilation factors of $\{1, 4, 9\}$ are employed to extract feature representations across multiple scales. Features

from these three branches are integrated via element-wise addition, followed by a SG and a SCA to further enhance feature representation. Finally, a gated MLP is employed to minimize information loss. Prior to different DBlock pairs, upsampling is performed using PixelShuffle, enabling the network to learn how to reconstruct fine-grained details.

Once the coherent signal covariance matrix is processed by the CMR-GAN generator, a full-rank covariance matrix is output. This generated matrix is concatenated along the first dimension with the real incoherent covariance matrix, resulting in a tensor of dimension $\{6, M, M\}$, which serves as the input to the discriminator. The discriminator is equipped with four convolution layers with varying channel counts of 16, 32, 64, and 128, where the kernel size, stride, and padding are consistently set to 4, 2, and 1, respectively. In the final stage, a sigmoid activation is employed to generate a probability score indicating whether the generated covariance matrix is full-rank or rank-deficient.

Since the objective of training a GAN is to render the discriminator unable to distinguish between generated and real samples, the convergence process differs from that of conventional neural networks. To ensure the stable convergence of the CMR-GAN, the aggregate generator loss is composed of a pixel-level reconstruction term and an adversarial component. The overall discriminator loss is computed as the arithmetic mean of the losses derived from classifying real and generated sample pairs. These two categories of losses can be formulated as follows

$$\begin{cases} \mathcal{L}_D = (\text{BCE}(\varphi_{\text{real}}, \hat{\varphi}_{\text{real}}) + \text{BCE}(\phi_{\text{fake}}, \hat{\phi}_{\text{fake}})) / 2B \\ \mathcal{L}_G = (\text{BCE}(\phi_{\text{fake}}, \hat{\phi}_{\text{fake}}) + \lambda \mathcal{L}_1(\mathbf{R}_{\text{pred}}, \mathbf{R}_{\text{incoh}})) / B, \end{cases} \quad (7)$$

where $\mathcal{L}_D$ and $\mathcal{L}_G$ denote the discriminator loss and generator loss, respectively. BCE and $\mathcal{L}_1$ represent the binary cross-entropy loss and the $\mathcal{L}_1$ loss, respectively. λ is the penalty term. φ_{real} , $\hat{\varphi}_{\text{real}}$, ϕ_{fake} , $\hat{\phi}_{\text{fake}}$, $\mathbf{R}_{\text{pred}}$, and $\mathbf{R}_{\text{incoh}}$ represent the real labels, predicted true labels, fake labels, predicted fake labels, reconstructed covariance matrix, and real incoherent covariance matrix, respectively. B denotes the batch size.

IV. SIMULATION RESULTS

This section assesses the efficacy of the proposed CMR-GAN. A ULA comprising $M = 16$ elements was adopted to acquire data for both training and validation purposes. In this setup, the number of snapshots L is maintained at 400, with SNR values spanning -10 dB to 20 dB at 2 dB intervals. The incident angles are confined to $[-60^\circ, 60^\circ]$. Adopting the angle combination strategy from [16], a total of 7,260 angle pairs are generated. For each pair, corresponding coherent and uncorrelated signals are generated to serve as the network input and the ground truth, respectively. The entire dataset is divided into training and validation subsets following an $8 : 2$ split. Regarding the generation of coherent signals, the amplitude fading coefficient is fixed at 1, while the phase difference followed a uniform distribution within $[0, 2\pi]$. The

performance estimation standard is the root mean square error (RMSE), defined as

$$\text{RMSE} = \sqrt{\frac{1}{MC \times K} \sum_{mc=1}^{MC} \sum_{k=1}^K (\hat{\theta}_{mc,k} - \theta_k)^2}, \quad (8)$$

where $\hat{\theta}_{mc,k}$ denotes the estimated value of the k -th source angle in the mc -th Monte Carlo simulation, θ_k is the true angle, K and MC represents the number of sources and Monte Carlo simulations, respectively.

Key hyperparameters for network training include a learning rate of 0.002, a penalty weight λ fixed at 30, and a batch size of 64 over 200 epochs. The ResNet model proposed in [13] and forward/backward spatial smoothing (FBSS-MUSIC) in [17] are selected as the baseline for comparison. To benchmark the theoretical performance limit, the Cramér-Rao Bound (CRB) for correlated signals is also plotted as a dashed reference. The MUSIC algorithm was employed as the subspace algorithm for subsequent DOA estimation for all models. Monte Carlo simulations are conducted 500 times to compare performance under different SNRs, snapshots, and numbers of coherent signals. All models are trained on a system equipped with dual Intel Xeon Scalable Platinum 8275CL CPUs and an NVIDIA RTX 4090 GPU. The software environment consists of Ubuntu 24.04, PyTorch 2.6, and Python 3.12.

For brevity, we append “-2” to the model name to indicate evaluation under two fully coherent signals. The same naming convention applies to the three-signal scenarios. Initially, the estimation performance is assessed across a spectrum of SNR levels, spanning -10 dB to 18 dB with a 4 dB interval. For this experiment, the snapshot number is maintained at 400 , and two coherent sources are positioned at -12.23° and 28.85° , which are chosen to avoid bias from grid-aligned estimators. As illustrated in Fig. 2, the RMSE of all models demonstrates a robust decreasing trend. However, CMR-GAN consistently achieves a lower RMSE than ResNet and FBSS-MUSIC at every SNR level. This indicates that CMR-GAN recovers the rank of the covariance matrix more effectively than two baseline models, and unlike FBSS-MUSIC, it does so without sacrificing array aperture. Subsequently, the number of coherent signals increases to three, with angles located at -32.77° , -12.23° , and 28.85° . As observed in Fig. 2, compared to the two-signal scenario, performances of CMR-GAN and FBSS-MUSIC degrade slightly, which is expected due to the increased difficulty, but still maintains a low RMSE. Conversely, ResNet fails to accurately estimate the angles in the presence of three coherent signals, suggesting an inability to correctly restore the rank of the covariance matrix, leading to estimation failure.

After that, experiments are conducted with different numbers of snapshots, specifically at five levels: 20 , 50 , 100 , 200 , and 300 . The SNR is maintained at 0 dB, with incident angles of -12.23° and 28.85° . The experimental findings derived from the simulation are depicted in Fig. 3. It is evident that the RMSE for all models exhibits a consistent downward trend as the snapshot count rises. CMR-GAN consistently outperforms

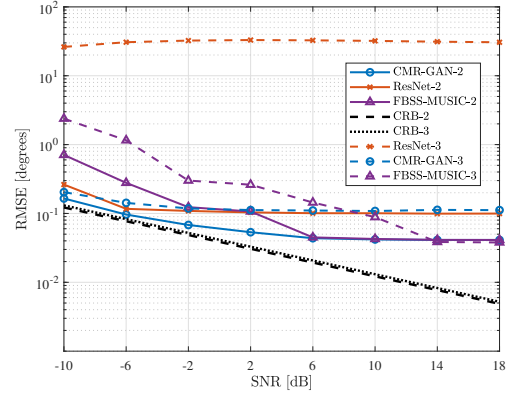


Fig. 2: RMSE versus SNRs in 2/3 coherent signals.

ResNet and FBSS-MUSIC, demonstrating its strong capability in feature reconstruction. Similar to the SNR experiments, when the number of incident signals is increased to three, ResNet fails to estimate the angles correctly. It is worth noting that even under low snapshot conditions, the performance of CMR-GAN-2 surpasses that of the ResNet baseline, further validating that CMR-GAN is better suited for reconstructing the covariance matrices of coherent signals.

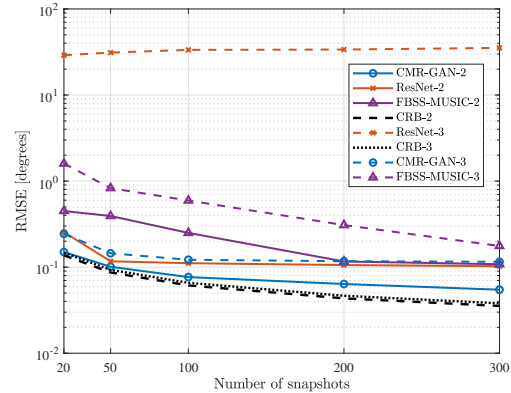


Fig. 3: RMSE versus snapshots in 2/3 coherent signals.

V. CONCLUSION

This work proposes an innovative deep learning architecture, namely CMR-GAN, designed to tackle the challenge of coherent signal DOA estimation. In contrast to conventional spatial smoothing methods that inevitably sacrifice the effective array aperture, our approach utilizes a GAN-based network to directly recover the full-rank covariance matrix from rank-deficient inputs. By incorporating an encoder-decoder architecture with SG and SCA, the proposed model effectively captures the underlying features of the signal subspace and reconstructs the covariance matrix. Simulation results show that CMR-GAN yields superior estimation accuracy and robustness compared to existing ResNet-based and spatial smoothing methods, particularly in challenging scenarios.

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